

POINT ASSIGNMENT INDEX

[illegible]

REV	DATE	DESCRIPTION	Created By	Checked by
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[illegible]

		DO		
Point Description				
Mean Well NDR-480-24 N (AC Neutral Input)	N	0	0	0
Mean Well NDR-480-24 G (Ground / PE)	G	0	0	0
Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Comn+V		0	0	0
Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit	+V	0	0	0
Mean Well NDR-480-24 +V to VFD FA (Jam Signal)	+V	0	0	0
Mean Well NDR-480-24 +V to F2-08AD-1 (Module Power)	+V	0	0	0
Mean Well NDR-480-24 +V to F2-08DA-2 (Module Power)	+V	0	0	0
Mean Well NDR-480-24 +V to F2-04RTD (Module Power)	+V	0	0	0
Mean Well NDR-480-24 +V to Indicator LEDs	+V	0	0	0
Mean Well NDR-480-24 -V to SD-N35 Contactor Coil A2	-V	0	0	0
Mean Well NDR-480-24 -V to D2-08ND3 (Input Common)	-V	0	0	0
Mean Well NDR-480-24 -V to D2-08TR (Relay Common)	-V	0	0	0
Mean Well NDR-480-24 -V to F2-08AD-1 (Module Power Return)	-V	0	0	0
Mean Well NDR-480-24 -V to F2-08DA-2 (Module Power Return)	-V	0	0	0
Mean Well NDR-480-24 -V to F2-04RTD (Module Power Return)	-V	0	0	0
Mean Well NDR-480-24 -V to VFD DCM (Digital Common)	-V	0	0	0
Mean Well NDR-480-24 -V to VFD ACM (Analog Common)	-V	0	0	0
Mean Well NDR-480-24 +V to HMI (EA1-T4CL)	+V	0	0	0
Mean Well NDR-480-24 -V to HMI (EA1-T4CL)	-V	0	0	0
Total Points		0	0	0

Destination Description

- 0 120 VAC line conductor from wall outlet, routed through panel main disco
- 0 1 AC Mai120 VAC neutral conductor from wall outlet, returns through neutral bus
- 0 1 Panel CEquipment grounding conductor; bonds PSU chassis to panel ground bus a
- 0 1 EURO Fused +24V distribution to operator pushbutton common via S4LH
- 0 1 Cutler- Direct +24V (no fuse) into E-stop safety chain
- 0 1 VFD HDirect +24V (no fuse) into VFD jam signal circuit
- 0 1 F2-08ADirect module power
- 0 1 F2-08DDirect module power
- 0 1 F2-04RDirect module power
- 0 1 LED incDirect +24V to (+) side of all illuminated pushbutton LED blocks and the se
- 0 1 Mitsubi Contactor coil return through E-stop chain
- 0 1 D2-08NModule input common for sourcing mode operation

in
common
reference for all 8 channels



Notes

Wall AC L -> main disconnect -> EURO S10H-5H 25 A slow-blow -> PSU L

Wall AC N -> neutral bus -> PSU N. White wire per NFPA 79.

PSU G stud -> panel ground bus. Green or Green/Yellow wire per NFPA 79.

+V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common si

+V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this li

+V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges ar

+V -> F2-08AD-1 +24V terminal. No fuse on this branch.

+V -> F2-08DA-2 +24V terminal. No fuse on this branch.

+V -> F2-04RTD +24V terminal. No fuse on this branch.

+V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), \

-V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop r

-V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH

-V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output termin

-V -> F2-08AD-1 0V terminal.

-V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog

-V -> F2-04RTD 0V terminal.

-V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/\

-V -> VFD ACM. Shared 0V reference for the analog speed command (VI fr

+V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (repl

-V -> HMI 0VDC. System DC common reference for the HMI panel.

. Black wire per NFPA 79.

).

side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3
re between E22B1 and A1 to monitor E-stop state.
rd delivers +24V to PLC X5.

Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when en
not pressed) and +V reaches A1.

I when +24V is applied.

ial connects to -V (sinks current from the load).

signal common.

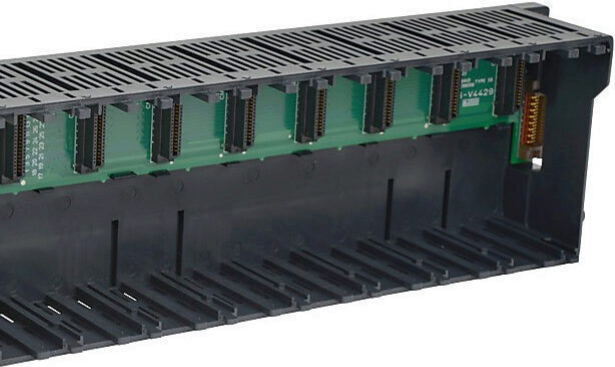
VPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM.
om F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).
aces former Rhino PSB12-030-P).

B x2, Reset S2PR-P3R).

ergized.

Direct Logic 205 D2-09B-1		POINT TYPE					
		Hardware Point					Virtual Point
Point Description	Origin Address	DO	DI	AO	AI	Pwr	
PLC Rack (DirectLOGIC 205) – Base	Base	0	0	0	0	1	—
PLC Rack (DirectLOGIC 205) – Line Input	L	0	0	0	0	1	—
PLC Rack (DirectLOGIC 205) – Neutral Input	N	0	0	0	0	1	—
PLC Rack (DirectLOGIC 205) – Ground (Protective Earth)	G	0	0	0	0	1	—
PLC Rack (DirectLOGIC 205) – Slot 1	Slot 1	0	0	0	0	0	—
PLC Rack (DirectLOGIC 205) – Slot 2	Slot 2	0	8				
PLC Rack (DirectLOGIC 205) – Slot 3	Slot 3	0	0				
PLC Rack (DirectLOGIC 205) – Slot 4	Slot 4	0	0				
PLC Rack (DirectLOGIC 205) – Slot 5	Slot 5	8	0				
PLC Rack (DirectLOGIC 205) – Slot 6	Slot 6	0	0				
PLC Rack (DirectLOGIC 205) – Slot 7	Slot 7	—	—				
PLC Rack (DirectLOGIC 205) – Slot 8	Slot 8	—	—				
PLC Rack (DirectLOGIC 205) – Slot 9	Slot 9	—	—				



Destination Address	Destination Description
D2-09B-1	Base / Power Unit
L	MORSETTITALIA EURO S10H-5H Fuse Block
N	Neutral Bus
G	Ground Bus
H2-DM1F	CPU / Ethernet Module
	t
	Current (A)
	:
	Voltage (V)

Notes
Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.
Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal
Neutral from wall -> directly connected to PLC N terminal
Protective earth -> connected to PLC ground terminal / chassis bond
CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.
8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors.
8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).
4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.
8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV).
8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).
Blank fill panel. No I/O. Slot reserved for future expansion.
Blank fill panel. No I/O. Slot reserved for future expansion.
Blank fill panel. No I/O. Slot reserved for future expansion.

DirectLOGIC PLC CPU(H2-DM1E)		POINT TYPE					
		Hardware Point					Virtua I Point
Point Description	Origin Address	DO	DI	AO	AI	Pwr	
Power and communication interface	Backplane Connector (Left Edge)	0	0	0	0	1	0
Ground reference	Backplane GND	0	0	0	0		
HMI communication port	CPU Comm Port	0	0	0	0		
Total Points		0	0	0	0		



Destination Address	Destination Description
D2-09B-1 Base Slot 1 Connector	Provides power and backplane communication
	Common ground for logic power
	4" C-More Micro-Graphic panel (operator interface)



Notes
—
—
CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel.

Output Module (D2-08TA)		POINT TYPE								
		Hardware Point					Virtu al Point			
Point Description	Origin Address	DO	DI	AO	AI	Pwr		Destination Address	Destination Description	Notes
Output Common (Hot)	C	1	0	0	0	1	0	120 V Hot Bus	ZipLink Terminal Block (ZL-RTB20-OUT). 1A inline slow blow fuse	120 V hot feed shared by all outputs
Output 0	OUT0	1	0	0	0	0	0	A1	TX Contactor (GH15-BN)	Drives TX coil via ZipLink
Output 1	OUT1	1	0	0	0	0	0	A1	BUS Contactor (GH15-BN)	Drives BUS coil via ZipLink
Output 2	OUT2	1	0	0	0	0	0	A1	F1 Contactor (GH15-BN)	Drives F1 coil via ZipLink
Output 3	OUT3	1	0						Thermal Overload Relay	Feeds NC contact 95 → 96 → F2 coil
Module Power Feed	Backplane	0	0						.20	No external wiring
	Total Points	5	0							



4" C-More Micro-Graphic HMI (EA1-T4CL)		POINT TYPE								
		Hardware Point					Virtu al Point			
Point Description	Origin Address	DO	DI	AO	AI	Pwr		Destination Address	Destination Description	Notes
HMI Serial Comm Port	COM Port	0	0	0	0	0	1	Serial Port	H2-DM1E	Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits
HMI Power +V	+VDC	0	0	0	0	1	0	+V	Mean Well NDR-480-24	HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail
HMI Power 0 V	0VDC	0	0	0	0	1	0	0V	Mean Well NDR-480-24	0 VDC return to Mean Well NDR-480-24 - V rail (system DC common)
HMI Ground	Ground	0	0	0	0	1	0	Ground	Ground	
	Total Points	0	0	0						



Output Module (F2-02DAS-2)		POINT TYPE								
		Hardware Point					Virtual Point			
Point Description	Origin Address	DO	DI	AO	AI	Pwr			Destination Address	Destination Description
Analog Common (CH1)	CH1-V	0	0	1	0	0	0	CM	VFD (GS1-20P2) through Ziplink ZL-RTB20	0V1 and CH1-V are internally shorted
Analog Output CH1	CH1+V	0	0	1	0	0	0	AL	VFD (GS1-20P2) though Ziplink ZL-RTB20	0–10 V speed command to VFD
Module Logic +24 V	+V1	0	0	0	0	1	0	VDC +24V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Powers analog output circuitry, also fused with 1A fast blow fuse
Module Logic 0 V	0V1	0	0	0	0	1	0	VDC 0V	PLC Base (D2-09B-1) through ZipLink ZL-RTB20	Common return for module logic
Slot Power Feed	Backplane	0	0							No external wiring
	Total Points	0	0							



Point Description	Origin Address	DO	DI	AO	AI
D2-08ND3 X0	X0	0	1	0	0
D2-08ND3 X1	X1	0	1	0	0
D2-08ND3 X2	X2	0	1	0	0
D2-08ND3 X3	X3	0	1	0	0
D2-08ND3 X4	X4	0	1	0	0
D2-08ND3 X5	X5	0	1	0	0
D2-08ND3 X6	X6	0	1	0	0
D2-08ND3 X7	X7	0	1	0	0
D2-08ND3 Input Common (C)	C	0	0	0	0
Total Points		0	8	0	0

Pwr Destination Address

0 Forward Pushbutton
0 Reverse Pushbutton
0 Run 1 / Deadman 1 Pushbutton
0 Run 2 / Deadman 2 Pushbutton
0 E-Stop Button Monitor
0 VFD Fault Relay
0 Reserved - Reset Pushbutton (planned)
0 Spare
1 Mean Well NDR-480-24 -V
1

Destination Description

Forward button S2PR-P3G with Contact block SA-CA with LED modu

Reverse button S2PR-P3Y with Contact block SA-CA with LED modul

Run 1 button S2PR-P3B with Contact block SA-CA with LED module

Run 2 button S2PR-P3B with Contact block SA-CA with LED module

E-stop button E22B1 with NC contact block, wired in series with Mitsul

VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC

Reset button S2PR-P3R with Contact block SA-CA with LED module

Unused

DC power supply 0V return rail. Both top-row and bottom-row C termin

Notes

Shared button +24V -> NO contact -> X0.

Shared button +24V -> NO contact -> X1.

Shared button +24V -> NO contact -> X2.

Shared button +24V -> NO contact -> X3.

PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V.

PSU +V -> VFD FA. VFD FC -> X5. FB unused.

Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).

Unwired.

C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A f

ast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the commor

1 +24V to X0/X1/X2/X3 (and X6 when wired).

Point Description	Origin Address	DO	DI	AO	AI
F2-08AD-1 +24V (Module Power)	+24V	0	0	0	0
F2-08AD-1 0V (Module Power Return)	0V	0	0	0	0
F2-08AD-1 CH1+ (Channel 1 Signal)	CH1+	0	0	0	1
F2-08AD-1 CH2	CH2	0	0	0	1
F2-08AD-1 CH3	CH3	0	0	0	1
F2-08AD-1 CH4	CH4	0	0	0	1
F2-08AD-1 CH5	CH5	0	0	0	1
F2-08AD-1 CH6	CH6	0	0	0	1
F2-08AD-1 CH7	CH7	0	0	0	1
F2-08AD-1 CH8	CH8	0	0	0	1
Total Points		0	0	0	8

Pwr Destination Address

1 Mean Well NDR-480-24 +V

1 Mean Well NDR-480-24 -V

0 VFD Motor Current Monitor

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

2

Destination Description

DC power supply +24V rail

DC power supply 0V return rail. Shared with VFD DCM and ACM (all t

VFD analog output Huanyang HY02D211B-T VO terminal, configured

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Notes

PSU +V -> Module +24V terminal directly. No fuse on this branch.

PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.

VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V).

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

Point Description	Origin Address	DO	DI	AO	AI
F2-04RTD +24V (Module Power)	+24V	0	0	0	0
F2-04RTD 0V (Module Power Return)	0V	0	0	0	0
F2-04RTD CH1	CH1	0	0	0	1
F2-04RTD CH2	CH2	0	0	0	1
F2-04RTD CH3	CH3	0	0	0	1
F2-04RTD CH4	CH4	0	0	0	1
Total Points		0	0	0	4

Pwr Destination Address

1 Mean Well NDR-480-24 +V

1 Mean Well NDR-480-24 -V

0 Spare

0 Spare

0 Spare

0 Spare

2

Destination Description

DC power supply +24V rail

DC power supply 0V return rail

Unused RTD input channel

Unused RTD input channel

Unused RTD input channel

Unused RTD input channel

Notes

PSU +V -> Module +24V terminal. Module installed in rack but no channels currently used in this build.

PSU -V -> Module 0V terminal.

Unwired. Available for future RTD temperature sensor.

Unwired.

Unwired.

Unwired.

Point Description	Origin Address	DO	DI	AO	AI
D2-08TR Y0	Y0	1	0	0	0
D2-08TR Y1	Y1	1	0	0	0
D2-08TR Y2	Y2	1	0	0	0
D2-08TR Y3	Y3	1	0	0	0
D2-08TR Y4	Y4	1	0	0	0
D2-08TR Y5	Y5	1	0	0	0
D2-08TR Y6	Y6	1	0	0	0
D2-08TR Y7	Y7	1	0	0	0
D2-08TR Relay Common (C)	C	0	0	0	0
Total Points		8	0	0	0

Pwr Destination Address

0 Forward Indicator LED
0 Reverse Indicator LED
0 Run 1 / Deadman 1 Indicator LED
0 Run 2 / Deadman 2 Indicator LED
0 Error / Fault Indicator
0 VFD Fault Reset
0 VFD Forward Run
0 VFD Reverse Run
1 Mean Well NDR-480-24 -V
1

Destination Description

LED module SA-LDG (green, 12-24 VDC) integrated in Forward pushb

LED module SA-LDY (yellow, 12-24 VDC) integrated in Reverse pushb

LED module SA-LDB (blue, 12-24 VDC) integrated in Run 1 pushbutt

LED module SA-LDB (blue, 12-24 VDC) integrated in Run 2 pushbutt

Wamco model 525 panel-mount LED indicator (28 VDC, 0.6 W)

Huanyang HY02D211B-T VFD RST terminal (multi-input, sinking/NPN

Huanyang HY02D211B-T VFD FOR terminal (multi-input, sinking/NPN

Huanyang HY02D211B-T VFD REV terminal (multi-input, sinking/NPN

DC power supply 0V return rail

Notes

PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0 contact. Relay sinks Y0 to -V when energized, lighting the LED.

PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1 contact. Relay sinks Y1 to -V when energized.

PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2 contact.

PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3 contact.

PSU +V -> Wamco 525 (+). Wamco 525 (-) -> Y4 contact. 28V-rated indicator slightly under-driven at 24V (ac

Y5 -> VFD RST. When Y5 energized, relay closes Y5 to common (PSU -V), pulling RST input to DCM. ~500 r

Y6 -> VFD FOR. When Y6 energized, pulls FOR to DCM. Forward shredding.

Y7 -> VFD REV. When Y7 energized, pulls REV to DCM. Used during PLC unjam routine.

C -> PSU -V. Sinking mode: when any Y0-Y7 closes, that terminal connects to -V.

ceptable, ~21 mA draw).

ns pulse during PLC unjam routine.

Point Description	Origin Address	DO	DI	AO	AI
F2-08DA-2 +24V (Module Power)	+24V	0	0	0	0
F2-08DA-2 0V (Module Power Return / Analog Commo)	0V	0	0	0	0
F2-08DA-2 +V1 (Channel 1 Signal)	+V1	0	0	1	0
F2-08DA-2 +V2	+V2	0	0	1	0
F2-08DA-2 +V3	+V3	0	0	1	0
F2-08DA-2 +V4	+V4	0	0	1	0
F2-08DA-2 +V5	+V5	0	0	1	0
F2-08DA-2 +V6	+V6	0	0	1	0
F2-08DA-2 +V7	+V7	0	0	1	0
F2-08DA-2 +V8	+V8	0	0	1	0
Total Points		0	0	8	0

Pwr Destination Address

1 Mean Well NDR-480-24 +V

1 Mean Well NDR-480-24 -V

0 VFD Analog Speed Reference

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

0 Spare

2

Destination Description

DC power supply +24V rail

DC power supply 0V return rail. Single 0V terminal serves both as pow

Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Unused channel

Notes

PSU +V -> Module +24V terminal directly. No fuse on this branch.

PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through

+V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

Unwired.

i the common 0V rail.

Point Description	Origin Address	DO	DI	AO	AI
VFD R (AC Line Hot Input)	R	0	0	0	0
VFD T (AC Neutral Input via Contactor and Breaker)	T	0	0	0	0
VFD Ground (PE)	Ground	0	0	0	0
VFD U (Motor Phase 1 Output)	U	0	0	0	0
VFD V (Motor Phase 2 Output)	V	0	0	0	0
VFD W (Motor Phase 3 Output)	W	0	0	0	0
VFD P+ (Brake Resistor +)	P+	0	0	0	0
VFD PR (Brake Resistor Return)	PR	0	0	0	0
VFD UPF	UPF	0	1	0	0
VFD DRV	DRV	0	1	0	0
VFD DCM (Digital Common, top row)	DCM	0	0	0	0
VFD SPL	SPL	0	1	0	0
VFD SPM	SPM	0	1	0	0
VFD SPH	SPH	0	1	0	0
VFD RST (Fault Reset Input)	RST	0	1	0	0
VFD REV (Reverse Run Input)	REV	0	1	0	0
VFD FOR (Forward Run Input)	FOR	0	1	0	0
VFD ACM (Analog Common, top row)	ACM	0	0	0	0
VFD VO (Analog Output, 0-10 V)	VO	0	0	1	0
VFD 10V	10V	0	0	1	0
VFD FA (Multi-Output 3 Relay NO)	FA	0	1	0	0
VFD FC (Multi-Output 3 Relay Common)	FC	0	1	0	0
VFD FB (Multi-Output 3 Relay NC)	FB	0	1	0	0
VFD 24V (Onboard 24V Supply)	24V	0	0	0	0
VFD DCM (bottom row)	DCM (bottom)	0	0	0	0
VFD 5V	5V	0	0	1	0
VFD ACM (bottom row)	ACM (bottom)	0	0	0	0
VFD AI (Analog Current Input, 4-20 mA)	AI	0	0	0	1
VFD VI (Analog Voltage Input, 0-10 V)	VI	0	0	0	1
VFD RS-	RS-	0	0	0	0
VFD RS+	RS+	0	0	0	0
Total Points		0	11	3	2

Pwr Destination Address

1 AC Mains Hot via main disconnect and 30 A breaker
1 AC Mains Neutral via Phoenix Contact TMC 8 3C 15A I
1 Panel Ground Bus
1 Banana Socket (Blue) -> Motor T1
1 Banana Socket (White) -> Motor T2
1 Banana Socket (Red) -> Motor T3
1 Brake Resistor ATO APCS-300R30-AD (+ terminal)
1 Brake Resistor ATO APCS-300R30-AD (return terminal)
0 Unused
0 Unused
1 Mean Well NDR-480-24 -V
0 Unused
0 Unused
0 Unused
0 D2-08TR Y5
0 D2-08TR Y7
0 D2-08TR Y6
1 Mean Well NDR-480-24 -V
0 F2-08AD-1 CH1+
0 Unused
0 Mean Well NDR-480-24 +V
0 D2-08ND3 X5
0 Unused
1 Unused
1 (same node as top-row DCM)
0 Unused
1 (same node as top-row ACM)
0 Unused
0 F2-08DA-2 +V1
1 Unused
1 Unused

Destination Description

120 VAC line conductor from wall outlet, routed through panel main dis
Neutral leg switched through the safety-disconnect contactor. As-bui
Equipment grounding conductor; bonds VFD chassis to panel ground
Panel-mount banana socket (blue, 4 mm) provides quick-disconnect
Panel-mount banana socket (white, 4 mm) provides quick-disconnec
Panel-mount banana socket (red, 4 mm) provides quick-disconnect
Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO mo
Dynamic braking resistor (same component as P+ row)
Multi-Output 2 (PD051), open-collector optocoupler
Multi-Output 1 (PD050), open-collector optocoupler
DC power supply 0V return - shared reference for FOR/REV/RST sink
Multi-Input 6 - default low-speed select
Multi-Input 5 - default middle-speed select
Multi-Input 4 - default high-speed select
PLC relay output (sinking NPN mode) driving fault reset
PLC relay output (sinking NPN mode) driving reverse run command
PLC relay output (sinking NPN mode) driving forward run command
DC power supply 0V return - shared reference for analog signals
PLC analog input module CH1, used as motor current monitor (PD054
+10 V reference output (for external potentiometer)
DC power supply +V rail
PLC discrete input X5 (jam signal)
Normally-closed contact - not used in active-high NO configuration
Onboard 24 VDC auxiliary supply, max 200 mA
Internally tied to top-row DCM
+5 V reference output
Internally tied to top-row ACM
Analog current input for speed reference. Speed command uses VI vo
PLC analog output module CH1, used as commanded frequency refer
RS-485 Modbus communication negative line
RS-485 Modbus communication positive line

Notes

Wall AC L -> disconnect -> 30 A breaker -> VFD R. Black wire per NFPA 79.

AS-BUILT: neutral leg is switched by the contactor instead of the hot leg. Note: NFPA 79 / NEC standard prac

VFD ground stud -> panel ground bus. Green or Green/Yellow wire.

VFD U -> blue banana socket -> motor terminal.

VFD V -> white banana socket -> motor terminal.

VFD W -> red banana socket -> motor terminal.

VFD P+ -> brake resistor terminal 1.

VFD PR -> brake resistor terminal 2.

Unwired.

Unwired.

VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.

Unwired.

Unwired.

Unwired.

PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.

PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.

PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.

VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out).

VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.

Unwired.

PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02).

VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).

Unwired.

Unwired. Panel uses external Mean Well NDR-480-24 instead.

Alternate landing point. Top-row DCM is used.

Unwired.

Alternate landing point. Top-row ACM is used.

Unwired.

F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.

Unwired. Modbus not used.

Unwired.

the purpose is to switch the HOT leg. Review for final report. When E-stop is pressed, SD-N35 coil drops -> contact o

pens -> VFD T loses neutral -> VFD shuts down.